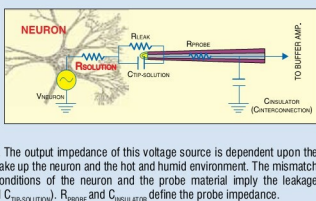


Neuron modeling

Neural signals have a voltage range of 100 μV to 700 μV and frequency range of 1 kHz to 3 kHz. The neuron can thus be viewed as a voltage source with a suitable voltage and frequency range

(V_{NEURON} in the figure). The output impedance of this voltage source is dependent upon the bio-chemicals that make up the neuron and the hot and humid environment. The mismatch and the boundary conditions of the neuron and the probe material imply the leakage impedance (R_{LEAK}), the tip-solution impedance ($R_{\text{TIP-SOLUTION}}$), the probe impedance (R_{PROBE}) and the capacitor impedance ($C_{\text{INSULATOR}}$) define the probe impedance.



to process neural signals to classify, characterise and identify them. It needs to analyse the operations of neural processes of several neurons (neural signals) simultaneously. So it needs a large number of inputs and outputs. A highly parallel architecture (a parallel architecture has multiple copies of a functional unit to process data simultaneously) is required to process such a large number of inputs.

The high traffic of neural data from the numerous probes necessitates its compression and therefore on-chip implementation of compression algorithms becomes necessary. Silicon-on-insulator (SOI) technology has a new substrate structure that has superior properties over the common bulk technology used to build the chips. It has greater noise immunity from cosmic radiation and can operate in low-voltage condition, making it a suitable option for building the implantable chips.

Neural signals. Neurons undergo continuous interactions in the brain. These have a prehensile structure with fingerlike projections at both the ends that serve as antennae (receiver and transmitter). Neurons do not physically touch each other and have a space between them called 'synapse.' Thousands of neurons may thus be connected to each other in virtually an infinite number of series and branching networks. Understanding the synaptic processes that gather, store and retrieve information throughout the brain is the

cutting edge of modern neuroscience.

EEG is a representation of the electrical activity of neurons in the brain. EEG signals are interesting artifacts and research is on to understand the origin of these signals. The electrical reaction of neurons is found to be of following types:

1. Presynaptic spike potential (rapid 1ms positive event resulting from presynaptic depolarisation)
2. Excitatory postsynaptic potential (prolonged 2ms graded positive potential)
3. Spike potential (high-voltage, sudden 2ms positive discharge of 10-30 mV)
4. After polarisation (prolonged positive potential)
5. Inhibitory postsynaptic potential (negative potential associated with neuronal inhibition)

Amplitudes of EEG signal voltage range from about 1 to 100 μV peak-to-peak at low frequencies (0.5 to 100 Hz) at the cranial surface. At the surface of the cerebrum, signals may be ten times stronger.

EEG signals are normally classified into five types:

1. *Delta*. 0.5-4Hz, less than 100 μV peak-to-peak
2. *Theta*. 4-8Hz, less than 100 μV peak-to-peak
3. *Alpha*. 8-13Hz, less than 10 μV peak-to-peak
4. *Beta*. 13-22Hz, less than 20 μV peak-to-peak

5. *Gamma*. 22Hz and higher, less than 2 μV peak-to-peak

Delta and theta signals are strongest over the central region and indicate sleep. These signals grow stronger with age and are almost undetectable in children. Alpha signals arise from the posterior brain in an awake person with closed eyes. Opening the eyes and focusing attention greatly reduces alpha waves. Beta activities are predominant over the central region at rest. High states of wakefulness and desynchronised alpha patterns produce beta waves. Gamma waves result from attention or sensory stimulation.

Following are the major areas of research on neural signals:

Spike sorting. Neurons generate electrical impulses (spikes) due to a phenomenon called 'action potential.' Spike sorting is grouping of these spikes based on the similarity of their shapes. Each spike train generated by a neuron has distinct characteristic. But amidst billion neurons, spikes interfere with each other and resultant spike trains become an ensemble from which identifying the components becomes difficult. In order to understand the association of a neuron to its spike train, it is essential to resolve the ensemble of neural signals. Spike sorting is thus a technique for classifying the neural signals based on their origin.

Neural signal processing. The vast amount of data associated with neural signal processing necessitates development of encoding of neural data for lesser bandwidth requirement. Lesser bandwidth alleviates several bottlenecks. First, it reduces the larger memory requirement in the chip, which directly impacts the chip-size, and a smaller chip size is always sought-after. Second, the processing speed improves because for the same operating frequency, the processor can process faster on a larger amount of data if the data size is small.

As mentioned earlier, neural signals are ensembles of spike trains. The recorded database of such ensembles is given a unique signature depending